Dynamic Sitemap System Documentation

🚀 Dynamic Sitemap Quick Start

✅ What's Done

Your site now has **automatic, self-updating sitemaps** that notify Google immediately when content is added.

🔗 Your New Sitemap URLs

```
https://orisuun.com/sitemap-events.xml
https://orisuun.com/sitemap-resources.xml
https://orisuun.com/sitemap-index.xml
```

📋 Next Steps (Do This Once)

1. Test the Sitemaps (5 minutes)

Open these in your browser: - https://orisuun.com/sitemap-events.xml - https://orisuun.com/sitemap-resources.xml

You should see XML with all your event/resource links.

2. Submit to Google Search Console (10 minutes)

1. Go to: https://search.google.com/search-console
2. Select your property (orisuun.com)
3. Click **Sitemaps** in left sidebar
4. Enter: `sitemap-index.xml`
5. Click **Submit**

3. Wait & Monitor

- Google starts crawling in **24-48 hours**
- Check **Coverage** report after 1 week
- All external links will be discovered automatically

🎯 How It Works Now

When Someone Adds an Event:

1. Form submits to database ✅
2. **Sitemap automatically updates** ✅
3. **Google is notified instantly** ✅
4. Google crawls new links within days ✅

When Someone Adds a Resource:

1. Form submits to database ✅
2. **Sitemap automatically updates** ✅
3. **Google is notified instantly** ✅
4. Google crawls new links within days ✅

No manual work required!

🧪 Quick Test

Add a Test Event:

1. Go to calendar page
2. Click "+ Add event"
3. Fill form and submit
4. Check console for: ✅ `Search engines notified successfully`
5. Visit `/sitemap-events.xml` - your event link should appear

🆘 Troubleshooting

Problem	Solution
Sitemap returns error	Check `.env` has `NEXT_PUBLIC_API_KEY`
Links not appearing	Wait 1 hour (cache refresh) or restart server

Problem	Solution
Google not crawling	Submit sitemap in Search Console
Notification fails	Non-critical - Google will still find updates

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📊 Expected Timeline

When	What Happens
Immediately	Sitemaps are live and working
1-2 days	Google starts crawling sitemaps
1 week	Links appear in Google's index
2-4 weeks	Full coverage of all links
Ongoing	Automatic updates forever

🎉 Benefits You Get

✅ All event/resource links discovered by Google ✅ Better SEO from outbound link recognition ✅ LLMs (ChatGPT, etc.) will see your links ✅ Zero maintenance - fully automatic ✅ Instant notifications on new content

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📞 Need Help?

1. Read full docs: `/docs/dynamic-sitemap-system.md`
2. Check Google Search Console for errors
3. View browser console for notification logs

Status: ✅ Deployed & Ready **Last Updated:** November 27, 2025 # Dynamic Sitemap System for SEO

Overview

This implementation creates **fully automatic dynamic sitemaps** that help Google and other search engines discover all external links from your calendar events and resources pages. The sitemaps update automatically whenever content is fetched, and search engines are notified immediately when new content is added.

🎯 What This Solves

Problem: Calendar events and resources are loaded dynamically via JavaScript, so their external links weren't being discovered by Google or LLMs.

Solution: Server-side generated sitemaps that include all external links, automatically updated, with instant notifications to search engines.

📁 Files Created

1. Sitemap Routes (Auto-updating)

- `/app/sitemap-events.xml/route.ts` - Dynamic sitemap for all calendar event links
- `/app/sitemap-resources.xml/route.ts` - Dynamic sitemap for all resource links
- `/app/sitemap-index.xml/route.ts` - Index that references all sitemaps

2. Search Engine Notification System

- `/app/api/sitemap/ping/route.ts` - API endpoint to ping Google/Bing
- `/utils/sitemapNotifier.ts` - Client-side utility functions

3. Updated Files

- `/app/(marketing)/calendar/popups/addevents.js` - Auto-notifies on event add
- `/app/(marketing)/calendar/popups/EditEventPopup.js` - Auto-notifies on event edit
- `/app/(marketing)/resources/components/resource-form.tsx` - Auto-notifies on resource add
- `/public/robots.txt` - Updated to reference all sitemaps

🚀 How It Works

Automatic Updates

1. **Dynamic Generation**: Sitemaps are generated on-demand from your database
2. **Fresh Data**: Every time Google requests a sitemap, it gets the latest data
3. **Caching**: Results are cached for 1 hour for performance
4. **Instant Notification**: When new content is added, Google is pinged immediately

When Google is Notified

✅ **Immediately after:** - Adding a new calendar event - Editing an existing event - Adding a new resource

What's Included in Sitemaps

Events Sitemap (`/sitemap-events.xml`): - Main calendar page - All external event links (EventLink field) - Only active, non-deleted events - Last modified dates for each link

Resources Sitemap (`/sitemap-resources.xml`): - Main resources page - All external resource links (Link/ResourceLink field) - Only active, non-deleted, non-paused, non-expired resources - Last modified dates for each link

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🔗 Sitemap URLs

Your site now has these sitemap endpoints:

```
https://orisuun.com/sitemap.xml              # Main site pages
https://orisuun.com/sitemap-events.xml       # Calendar events
https://orisuun.com/sitemap-resources.xml    # Resources
https://orisuun.com/sitemap-index.xml        # Index of all sitemaps
```

📊 Testing & Verification

1. Test Sitemap Generation

Visit these URLs in your browser:

```
https://orisuun.com/sitemap-events.xml
https://orisuun.com/sitemap-resources.xml
```

You should see XML with all your external links listed.

2. Verify Search Engine Notification

After adding an event or resource, check your browser console for:

✅ `Search engines notified successfully`

3. Submit to Google Search Console

1. Go to Google Search Console
2. Navigate to **Sitemaps**

3. Add these sitemaps:
   - `sitemap-index.xml` (recommended - includes all others)
   - OR individually: `sitemap-events.xml` and `sitemap-resources.xml`
4. Click **Submit**

Google will start crawling within 24-48 hours.

🛠️ Manual Operations

Manually Notify Search Engines

If you do bulk operations or want to manually trigger notifications:

```
import { notifySearchEngines } from '@/utils/sitemapNotifier';

// After bulk event updates
await notifySearchEngines('event');

// After bulk resource updates
await notifySearchEngines('resource');

// After bulk updates to both
await notifySearchEngines('all');
```

API Endpoint for Manual Pinging

You can also trigger notifications via HTTP:

```
# Ping for events
curl -X POST https://orisuun.com/api/sitemap/ping \
  -H "Content-Type: application/json" \
  -d '{"type": "event"}'

# Ping for resources
curl -X POST https://orisuun.com/api/sitemap/ping \
  -H "Content-Type: application/json" \
  -d '{"type": "resource"}'

# Ping for all
curl -X POST https://orisuun.com/api/sitemap/ping \
  -H "Content-Type: application/json" \
  -d '{"type": "all"}'
```

🔄 Update Frequency

Component	Update Frequency	Cached For
Sitemap Generation	Every request	1 hour
Search Engine Ping	Immediate on content add/edit	N/A
Google Crawl	1-7 days (Google's schedule)	N/A

📈 Expected Results

Short-term (1-2 weeks)

- Google will discover and index the sitemaps
- External links will start appearing in Google's index
- You can verify in Search Console → Coverage report

Long-term (1-3 months)

- Improved crawl coverage of external sites
- Better link discovery for SEO
- LLMs will see these links when training/updating
- Increased referral traffic potential

🧪 Monitoring & Maintenance

Check Sitemap Health

Google Search Console: - Navigate to **Sitemaps** section - View submitted sitemaps - Check for errors or warnings

Monitor Logs: - Server logs will show sitemap requests - Console logs show notification success/failure

Common Issues & Solutions

Issue: Sitemap returns empty or error - **Solution:** Check API credentials in `.env` - Verify backend API is accessible - Check console for error messages

Issue: Google not crawling sitemap - **Solution:** Manually submit in Search Console - Check robots.txt is accessible - Verify sitemap XML is valid

Issue: Notifications failing - **Solution:** Non-critical - sitemaps still work - Google will discover updates on next scheduled crawl - Check network connectivity

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🔐 Security & Performance

Performance Optimizations

✅ 1-hour caching reduces database load ✅ Non-blocking notifications (won't slow down forms) ✅ Fallback sitemaps on API errors ✅ Efficient XML generation

Security

✅ Uses existing API authentication ✅ No sensitive data exposed ✅ Public endpoints (sitemaps are meant to be public) ✅ Rate limiting via Next.js built-in caching

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🎉 What You Get

✅ **Automatic sitemap updates** - No manual work required ✅ **Instant Google notifications** - Faster indexing ✅ **Better SEO** - All external links discovered ✅ **LLM visibility** - AI models will see your links ✅ **Easy monitoring** - Google Search Console integration ✅ **Zero maintenance** - Works automatically forever

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🚨 Important Notes

1. **First-time setup**: Submit sitemaps to Google Search Console after deployment
2. **Testing**: Wait 24-48 hours for Google to start crawling
3. **Monitoring**: Check Search Console weekly for the first month
4. **Staging**: Test on staging environment first
5. **API Keys**: Ensure `NEXT_PUBLIC_API_KEY` is set in `.env`

📞 Support

If you encounter issues: 1. Check browser console for errors 2. Verify sitemap XML loads correctly 3. Check API responses in Network tab 4. Review Google Search Console for errors

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🎓 Learn More

- Google Sitemap Protocol
- Google Search Console Help

- Next.js Dynamic Routes

Last Updated: November 27, 2025 **Status:** ✅ Production Ready